

SAA: SCALE-AWARE ATTENTION BLOCK FOR MULTI-LESION SEGMENTATION OF FUNDUS IMAGES

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ABSTRACT

Multiple lesion segmentation, namely the segmentation of microaneurysms, soft exudate, hard exudate, and haemorrhage is very important to diabetic retinopathy diagnosis. However, the scales of different kinds of lesions are inconsistent. This inconsistent scale problem is unavoidable in the unified architecture design in which identical time of downsampling operations is used for different kinds of lesions. To achieve better performance at different scales, multiscale features need to be captured and adjusted. In this paper, we simply consider features from different stages of an encoder-decoder network as multiscale features. To re-weight importance of multiscale features dynamically, a scale-aware attention (SAA) block which consists of a spatial path and a channel path is introduced. In SAA block, adjusting operations are performed scale-wise instead of channel-wise or uniformly for all scales. Extensive experiments were conducted on two publicly-available datasets to verify the effect of SAA. SAA surpasses popular attention blocks and state-of-the-art results in the overall evaluation while comparable performance can be achieved in the individual evaluation at the same time.

Index Terms— Lesion segmentation, inconsistent scale, encoder-decoder, attention block

1. INTRODUCTION

Diabetic retinopathy (DR) is a vascular disease that affects normal blood vessels in the eye. It is the leading cause of preventable blindness worldwide[1]. According to the World Vision Report¹ released by the World Health Organization in October 2019, there are 146 million people worldwide suffering from DR. Retinal lesions, including microaneurysms (MA), soft exudate (SE), hard exudate (EX), and haemorrhage (HE), are closely related to the diagnosis of DR. Ophthalmologists usually diagnose DR based on the appearance of these retinal lesions. Therefore, performing retinal lesion segmentation is of great importance.

There exists a inconsistent scale problem in multiple lesion segmentation. From Fig.1 we can see that the four kinds of lesions vary in scale. Even within a certain kind, the scale

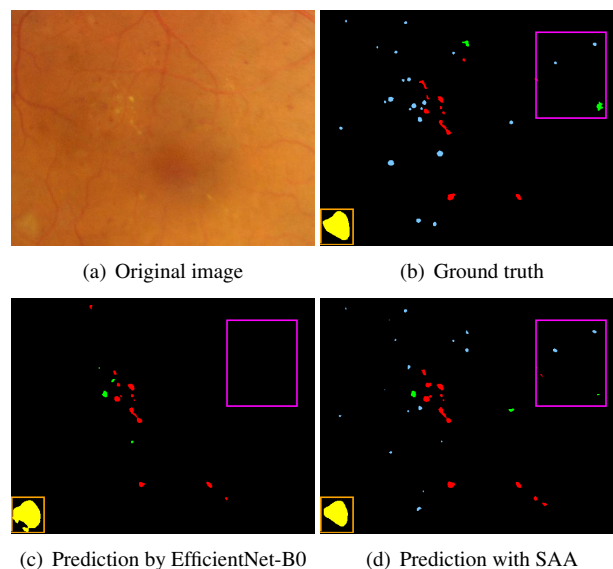


Fig. 1. Case study on IDRiD. Red represents for EX, green for HE, yellow for SE and blue for MA. Different bounding boxes highlight the comparison areas. Better viewed in color.

is inconsistent, e.g. EX. In separate lesion segmentation, the inconsistent scale problem can be solved by utilizing different architecture. We can observe from a recent review[2] that many works used distinctive structure with different time of downsampling operation for specific lesions. However, in our comprehensive design, the architecture is unified for different kind of lesions. There ought to be other methods to solve the inconsistent scale problem. In this paper, a simple yet effective scale-aware attention (SAA) block is proposed to dynamically adjust multiscale features. The multiscale features are re-weighted scale-wise in both spatial and channel path.

Our main contributions are as following,

- We utilize a unified architecture which can segment four kinds of lesions simultaneously. When be compared with state-of-the-art, its performance is better in overall evaluation and comparable in individual evaluations.
- We propose a scale-aware attention block aiming to solve the inconsistent scale problem. SAA block can achieve better results than other popular attention blocks such as

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¹<https://www.who.int/publications/i/item/world-report-on-vision>

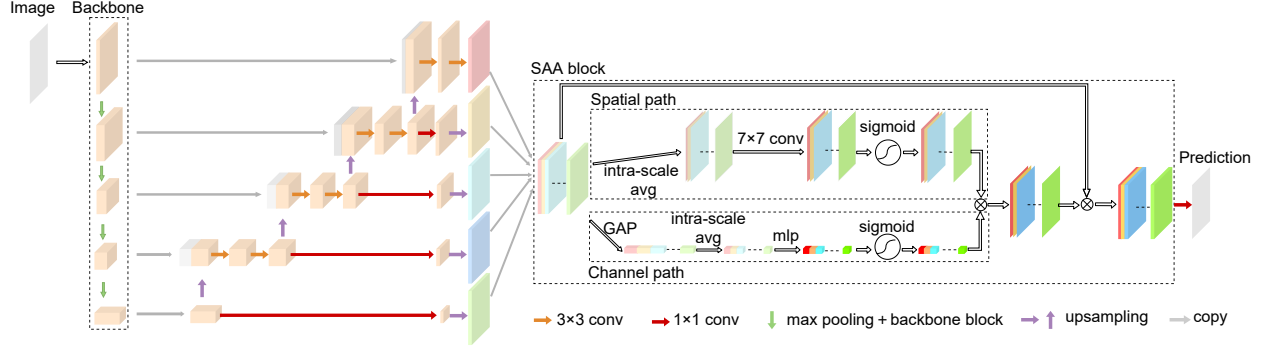


Fig. 2. The network is built on encoder-decoder architecture. And SAA block consists of a spatial path and a channel path.

SE block[3], CBAM[4] and CAB[5].

- SAA is a universal block which can be plugged into any backbone. And extensive experiments were conducted to verify the effect of SAA on different backbones. We will release our code at <https://github.com/nkicls/SAA>.

2. METHOD

As shown in Fig.2, the framework is designed in an end-to-end manner, which can output segmentation results of four kinds of lesions simultaneously in one inference. In this section, we will first introduce the overall architecture. Then the proposed SAA block will be shown in detail.

2.1. Overall architecture

As shown in Fig. 2, the architecture consists of the following parts:

Encoder-decoder architecture: The encoder-decoder structure has been widely used in medical image segmentation. Here we also adopt this structure. However, feature extraction ability of vanilla U-Net[6] is inadequate. So we replace the encoder portion with backbones which have better feature learning ability.

Multiscale feature shortcut: There exist a lot of ways to extract multiscale features, including ASPP[7], pyramid pooling[8], etc. However, the encoder-decoder structure itself already contains sufficient multiscale information. Here we simply consider features from different stages of the decoder as multiscale features. To maintain the balance between different scales, we first use a 1×1 convolution to reduce the dimension of features from coarser stages to the same as the finest scale. Bilinear interpolation is performed after 1×1 convolution to make coarser features the same scale as label. Using multi-scale shortcuts can provide direct paths between the final inference layers and the earlier layers, which can alleviate the problem of vanishing gradient.

Attention feature fusion: After the above operation, we can get several multiscale features. Features from different

scales play different roles for the segmentation of different lesions. The importance of multiscale features needs to be adjusted dynamically. To this end, attention blocks are appropriate here.

2.2. Scale-Aware Attention block

The proposed Scale-Aware Attention (SAA) block is shown in Fig.2. It composes of two paths, namely spatial path and channel path. In each path, unlike CBAM, only one branch is used. Given input features F_{in} consisting of S scales, each scale can be denoted as $F_{in}^i \in R^{H \times W \times C}$, $i \in \{1, 2, \dots, S\}$, where H , W and C denote height, width and number of channels of the feature maps, respectively. In spatial path, a cross channel average operation is first performed for each scale:

$$F_{spa.avg}^i = \frac{1}{C} \sum_{j=1}^C F_{in}^{i,j}, i \in \{1, 2, \dots, S\} \quad (1)$$

$F_{in}^{i,j}$ denotes the j -th feature of F_{in}^i . The resulting $F_{spa.avg}^i$ of every scale s are then concatenated together to feed to a series of operations including a 7×7 convolution and a sigmoid activation to learn scale-specific importance for every pixel in spatial dimension. This operation can be denoted as:

$$F_{spa.att} = \sigma(\text{conv}_{7 \times 7}([F_{spa.avg}^1, F_{spa.avg}^2, \dots, F_{spa.avg}^S])) \quad (2)$$

In channel path, global average pooling(GAP) operation is conducted first to get global features, which can be denoted as $F_{global}^i \in R^{1 \times 1 \times C}$, $i \in \{1, 2, \dots, S\}$. Then an intra-scale channel operation is conducted:

$$F_{cha.avg}^i = \frac{1}{C} \sum_{j=1}^C F_{global}^{i,j}, i \in \{1, 2, \dots, S\} \quad (3)$$

$F_{global}^{i,j}$ denotes the j -th node of F_{global}^i . The resulting $F_{cha.avg}^i$ are fed to a mlp layer and then a sigmoid activation to get channel-wise importance for each scale:

$$F_{cha.att} = \sigma(\text{mlp}([F_{cha.avg}^1, F_{cha.avg}^2, \dots, F_{cha.avg}^S])) \quad (4)$$

The pixelwise product of F_{spa_att} and F_{cha_att} is the weight of each scale, which can be denoted as F_{att} . Finally, the activated feature at the i -th scale can be got through pixel-wise product of F_{in}^i and F_{att}^i . This step can be denoted as:

$$F_{out}^i = F_{spa_att}^i \otimes F_{cha_att}^i \otimes F_{in}^i \quad (5)$$

The superscript i denotes the i -th scale and $\otimes$ denotes element-wise multiplication.

Activation weights of multiscale features are learned scale-wise in channel path of SAA, unlike SE block and CBAM. In channel dimension, the adjusting weights of features in a specific scale are totally the same. In spatial path, CBAM only gets a single activation map for features of all scales. In comparison, SAA gets S activation maps, each representing the importance for a specific scale in spatial dimension. The whole design of SAA can capture scale-wise importance of multiscale features, which is consistent with our goal to solve the problem of inconsistent scale.

3. EXPERIMENTS

3.1. Experiment implementation

Datasets: We evaluated our method on two publicly available datasets, namely IDRiD[9] and DDR[10]. The Indian Diabetic Retinopathy Image Dataset (IDRiD) dataset is the official dataset of the grand challenge ‘‘Diabetic Retinopathy – Segmentation and Grading’’ held by ISBI in 2018. We only used the segmentation part of IDRiD dataset, which consists of 81 images with pixel-level annotation. Among the dataset, all the images have annotations of EX and MA, 80 have HE and 40 have SE. The dataset was split into 54 training images and 27 testing images, the same as teams participating in the challenge. The DDR dataset consists of 13,673 images collected from 147 hospitals. 757 images of DDR contain pixel-level annotation, among which 601 images have HE, 570 images have MA, 239 images have SE and 486 images have EX. It has already been officially split into 383 training images, 149 validation images and 225 test images.

Evaluation metric: We used Area Under Precision Recall curve (AUPR) as the evaluation metric, same as the one used in the ISBI2018 challenge. We also utilized mean AUPR (mAUPR) to evaluate the overall performance on multi-lesion segmentation. It is calculated by the average of AUPR of all the four kinds of lesions, like the definition of mAP in object detection. The mAUPR metric was also used in a state-of-the-art work L-seg[11].

Implementation details: The original resolution of images from IDRiD is 4288×2848, which is relatively large for our GPU. So we first resized images and their corresponding labels to 1440×960. We also conducted data augmentation by flipping horizontally and vertically, and rotation by an angle of 90, 180 and 270. For the pre-processing of DDR, we cropped the original images into squares and resized them to

Table 1. Ablation study on IDRiD dataset illustrating the role of the two paths.

Method	EX	HE	SE	MA	mAUPR
baseline	0.8718	0.6097	0.6681	0.3098	0.6148
+ only shortcut	0.8738	0.6006	0.6784	0.3217	0.6186
+ channel path	0.8779	0.6079	0.6687	0.3907	0.6363
+ spatial path	0.8792	0.6358	0.6785	0.3793	0.6432
+ 2 paths (SAA)	0.8766	0.6388	0.6901	0.3979	0.6509

1024×1024. We did not conduct data augmentation for DDR. Backbones we used include MobileNet[12], EfficientNet[13] and MixNet[14], which were all pretrained on DDR dataset previously. The initial learning rate was set to 0.001 and adjustment strategy adopted was poly learning rate decay. The power of poly strategy was set to 0.9. All the experiments were trained for 60 epochs using cross entropy loss. We adopted Adam as the optimizer. All the experiments were performed on a single Nvidia RTX3090 GPU with 24 GB of memory.

3.2. Results

3.2.1. Ablation studies on IDRiD dataset

We conducted ablation studies on IDRiD dataset, which consists of two parts as shown in the following subsection.

Analysis of SAA block: We first analyze the role of the two paths of SAA block. In this part, experiments conducted include baseline, baseline with spatial or channel path only and baseline with the two paths together, namely the proposed SAA block. The backbone used here is MobileNetV3-Large. The result shown in Tab.1 illustrates that adding shortcuts only achieves a slight improvement compared to the baseline, which means the multiscale features need to be adjusted in a better way. Model with attention along either path achieves better results than the baseline. And the model with SAA block achieves the best result in the overall mAUPR and individual evaluation of HE, SE and MA. The experiments illustrate that both the spatial path and channel path are effective for re-weighting multiscale features, and the best results can be achieved with the two paths together.

Comparison with other attention blocks: We conducted experiments on several popular backbones including MixNet-M, MobileNetV3-L and EfficientNet-B0 to verify the versatility of SAA. We compare SAA with widely used attention blocks including SE block, CBAM and CAB. The results shown in Tab.2 illustrate that in the overall evaluation of mAUPR, SAA surpasses SE block, CBAM and CAB on all the backbones. SAA also ranks first in six of twelve individual evaluations and second in three. We can conclude that compared with other three popular attention blocks, SAA is more suitable to solve the inconsistent scale problem.

Table 2. Comparison of SAA with other popular attention blocks on several backbones on IDRiD dataset.

Backbone	Method	EX	HE	SE	MA	mAUPR
MobileNetV3-L	baseline	0.8718	0.6097	0.6681	0.3098	0.6148
	+SEblock	0.8778	0.6178	0.6447	0.3796	0.6300
	+CBAM	0.8822	0.5946	0.6625	0.3778	0.6293
	+CAB	0.8747	0.5947	0.7291	0.3782	0.6442
	+SAA	0.8766	0.6388	0.6901	0.3979	0.6509
MixNet-M	baseline	0.8738	0.6275	0.6516	0.3339	0.6217
	+SEblock	0.8813	0.6359	0.6829	0.3254	0.6314
	+CBAM	0.8574	0.6535	0.6060	0.3885	0.6264
	+CAB	0.8784	0.6018	0.6793	0.3635	0.6308
	+SAA	0.8765	0.6353	0.7010	0.3773	0.6475
EfficientNet-B0	baseline	0.8755	0.6229	0.6743	0.3514	0.6310
	+SEblock	0.8784	0.6473	0.6876	0.4152	0.6571
	+CBAM	0.8838	0.6335	0.6328	0.3992	0.6373
	+CAB	0.8805	0.6502	0.7025	0.4134	0.6617
	+SAA	0.8812	0.6704	0.7281	0.4152	0.6738

Table 3. Experiments on DDR illustrating the effect of SAA.

Backbone	Method	EX	HE	SE	MA	mAUPR
MobileNetV3-L	baseline	0.6172	0.4023	0.2301	0.1857	0.3588
	+SAA	0.6221	0.3706	0.2854	0.1852	0.3658
MixNet-M	baseline	0.6376	0.4258	0.2167	0.1788	0.3647
	+SAA	0.6422	0.4114	0.2887	0.1910	0.3833
EfficientNet-B0	baseline	0.6318	0.4504	0.2680	0.1900	0.3851
	+SAA	0.6269	0.4456	0.3747	0.1933	0.4102

3.2.2. Experiments on DDR dataset

We also conducted experiments on DDR dataset to show the impact of introducing SAA to baseline. In this part, experiments conducted include baselines and models with SAA block. The baselines utilized are the same as ones used in ablation studies on IDRiD dataset. The results are shown in Tab.3. From the table we can see that all the three models with SAA achieve the best result on mAUPR. In individual evaluation, there is a huge improvement on SE. In the evaluation of other three lesions, the performance with SAA is comparable with or slightly surpasses the baselines. The experiment results illustrate that SAA also works well on DDR dataset.

3.2.3. Comparison with state-of-the-art results

In this part, comparisons with state-of-the-art results on both IDRiD dataset and DDR dataset are illustrated. The comparison is shown in Tab.4. The first three columns of IDRiD part are the top three teams from the IDRiD2018 challenge. From the table we can see that EfficientNet-B0 with SAA surpass all the results in the overall evaluation of mAUPR, which is consistent with our initial goal. In the individual evaluation,

Table 4. Comparison with state-of-the-art results on IDRiD and DDR datasets.

Method	EX	HE	SE	MA	mAUPR
IDRiD dataset					
VRT (1st)[9]	0.7127	0.6804	0.6995	0.4951	0.6469
PATech (2nd)[9]	0.8850	0.6490	–	0.474	–
iFLYTEK-MIG (3rd)[9]	0.8741	0.5588	0.6588	0.5017	0.6484
L-seg[11]	0.7945	0.6374	0.7113	0.4627	0.6515
MobileNetV3-L+SAA	0.8766	0.6388	0.6901	0.3979	0.6509
MixNet-M+SAA	0.8765	0.6353	0.7010	0.3773	0.6475
EfficientNet-B0+SAA	0.8812	0.6704	0.7281	0.4152	0.6738
DDR dataset					
U-Net[6]	0.5639	0.3331	0.0966	0.1064	0.2750
U-Net++[15]	0.5087	0.2439	0.0973	0.1370	0.2467
L-seg[11]	0.5546	0.3586	0.2648	0.1052	0.3208
MobileNetV3-L+SAA	0.6221	0.3706	0.2854	0.1852	0.3658
MixNet-M+SAA	0.6422	0.4114	0.2887	0.1910	0.3833
EfficientNet-B0+SAA	0.6269	0.4456	0.3747	0.1933	0.4102

we achieve the best performance on SE segmentation. In the evaluation of EX and HE, EfficientNet-B0 with SAA achieve the second best result. Note that our result on MA is not good enough. It is because that the architecture we adopted used four times of downsampling operation to accomplish the trade-off between overall and individual evaluation. As for the DDR dataset, huge improvements in each evaluation can be observed compared with two popular models and L-seg.

3.3. Visualization

To understand the effect of SAA block more intuitively, a real case from IDRiD dataset is shown in Fig.1. Backbone used here is EfficientNet-B0. The orange bounding box illustrates that the false positive irregular area of SE does not appear in models with SAA as in the baseline. It is very clear to see that more small lesions are recognized in models with SAA in the pink bounding box. We can also observe appearance of more MAs in other areas of the figure. From the case shown we can know that SAA works well both on small and large lesions.

4. CONCLUSION

In this paper, we utilized a unified structure to perform multi-lesion segmentation simultaneously. To overcome the inconsistent scale problem, we designed a scale-aware attention block to adjust features scale-wise. Extensive experiments have shown that SAA outperforms popular attention blocks and sota works in the overall evaluation when comparable performance can be achieved in individual evaluations. SAA block is not just designed for lesion segmentation on fundus images. We believe it can also work well in other situations which also have the inconsistent scale problem.

5. REFERENCES

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